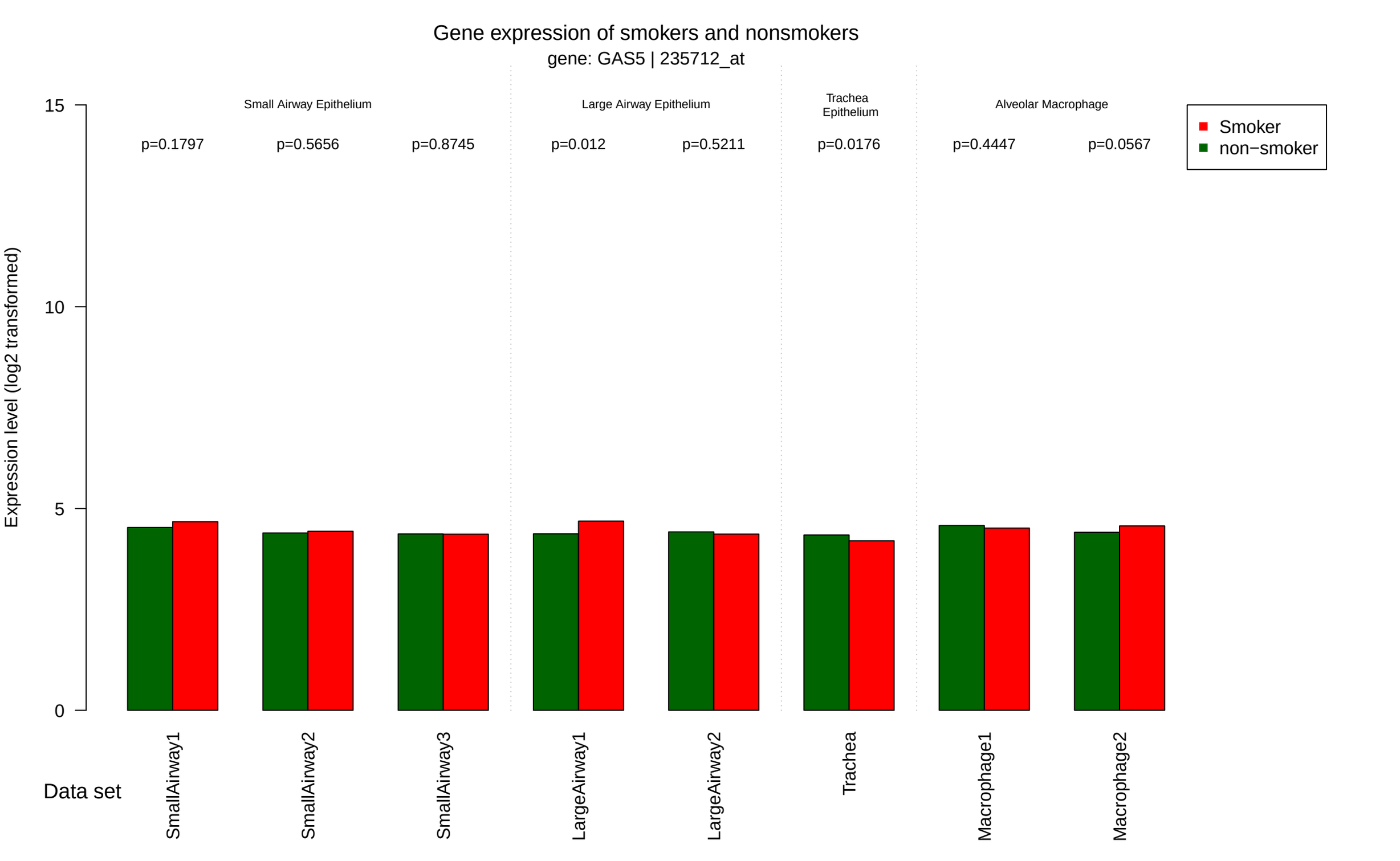
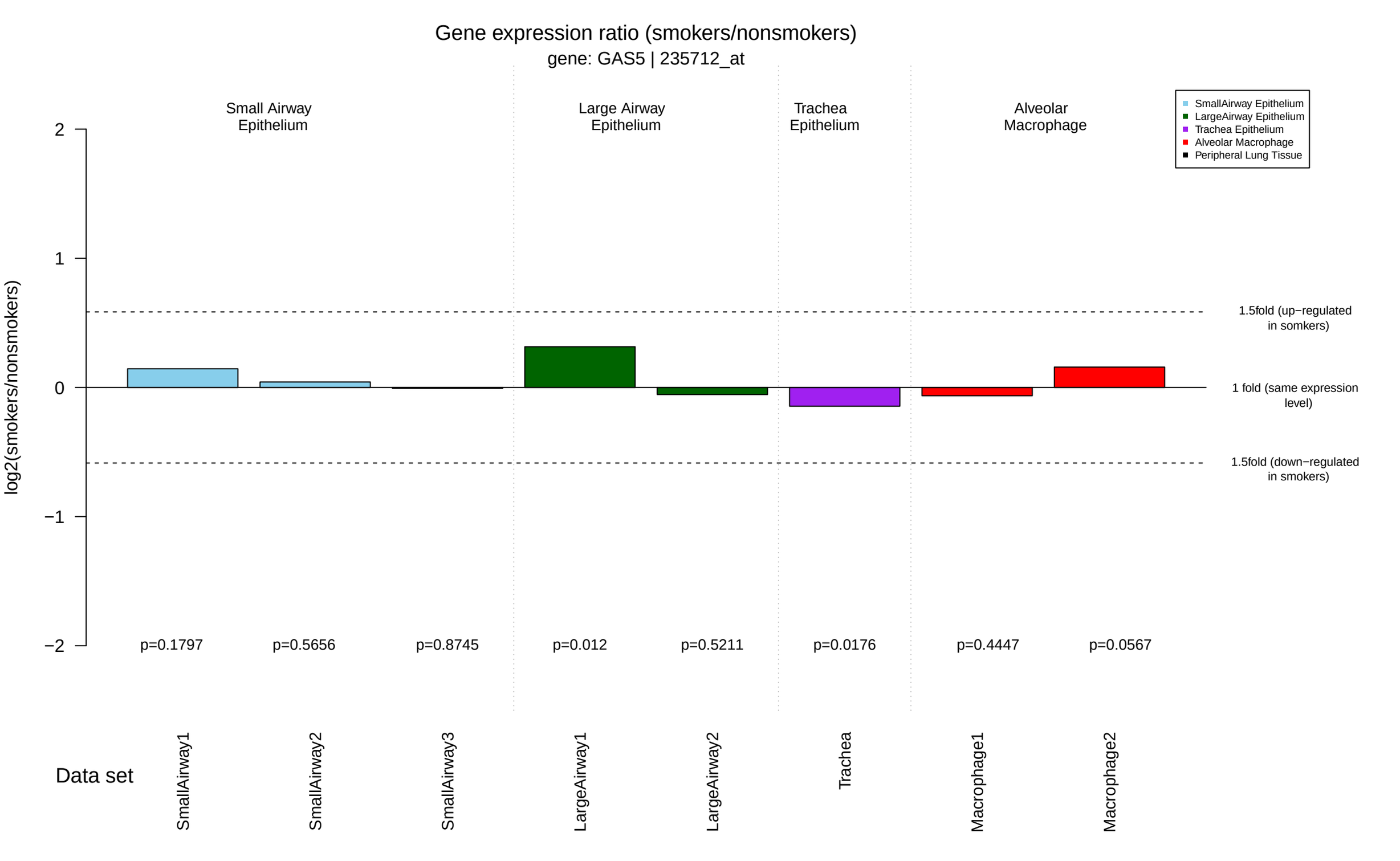
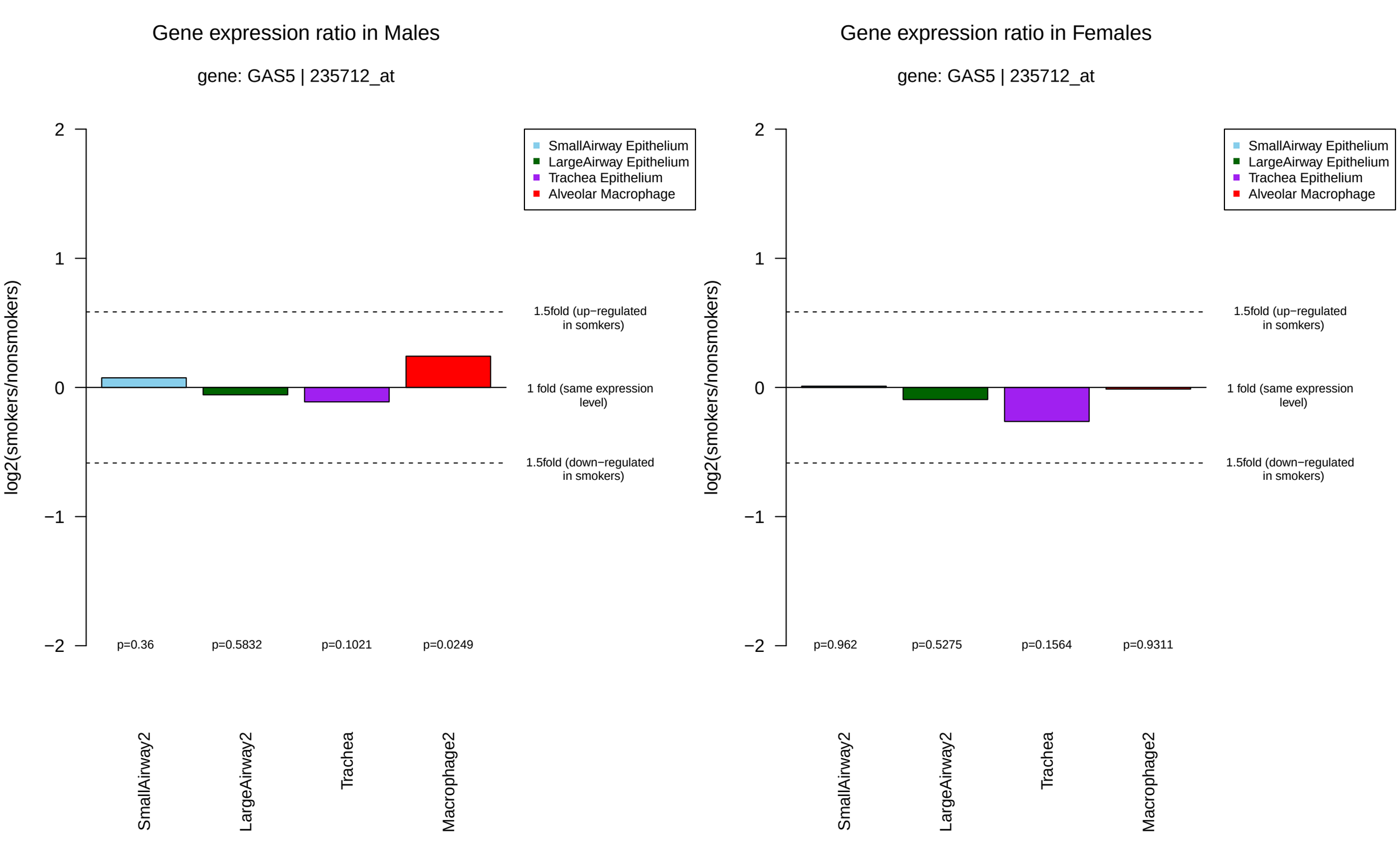
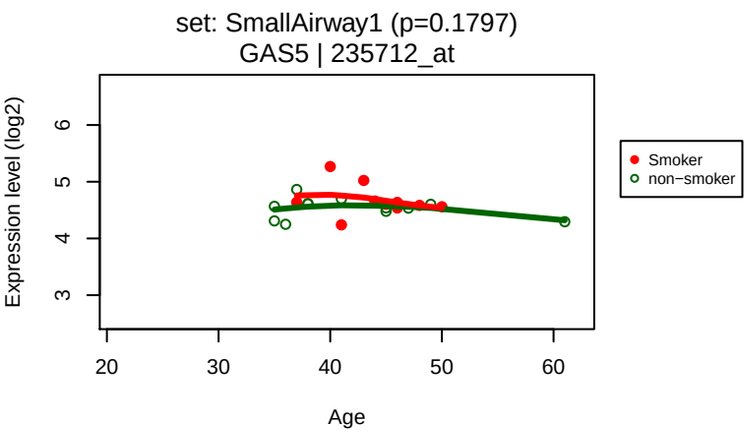
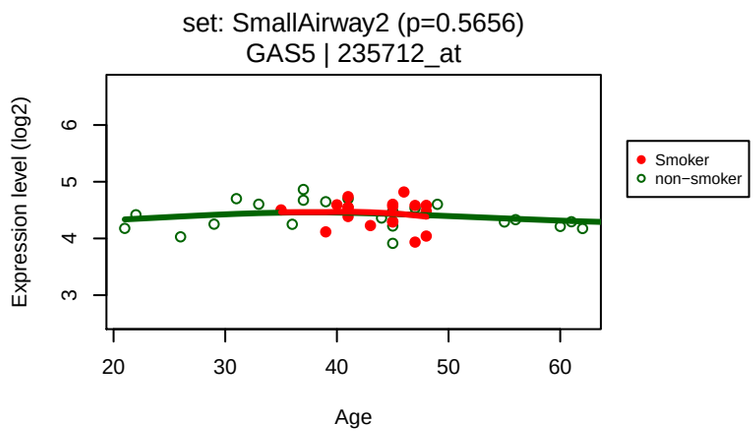
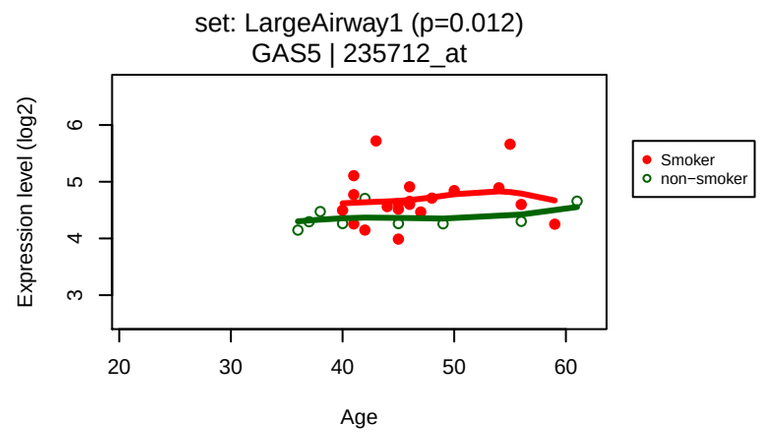

Gene expression of smokers and nonsmokers
gene: GAS5 | 235712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1797
p=0.5656
p=0.8745
p=0.012
p=0.5211
p=0.0176
p=0.4447
p=0.0567
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

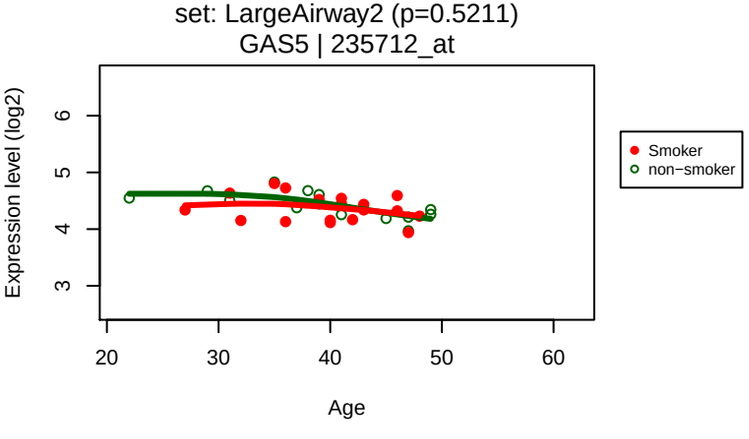

Gene expression ratio (smokers/nonsmokers)
gene: GAS5 | 235712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1797
p=0.5656
p=0.8745
p=0.012
p=0.5211
p=0.0176
p=0.4447
p=0.0567
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

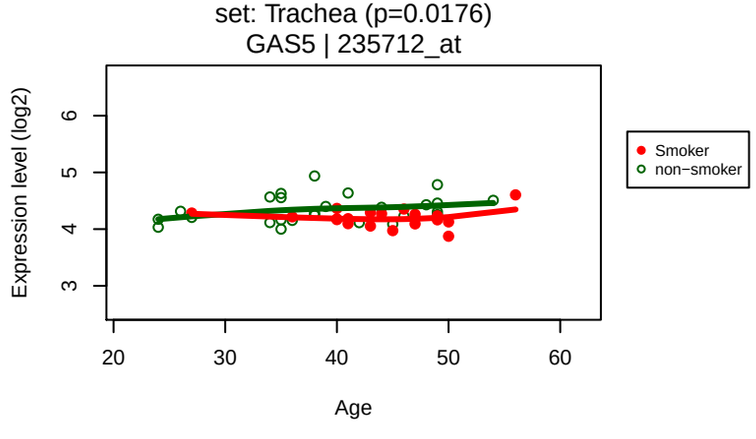

Gene expression ratio in Males
gene: GAS5 | 235712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.36
p=0.5832
p=0.1021
p=0.0249
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: GAS5 | 235712_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.962
p=0.5275
p=0.1564
p=0.9311
SmallAirway2
LargeAirway2
Trachea
Macrophage2

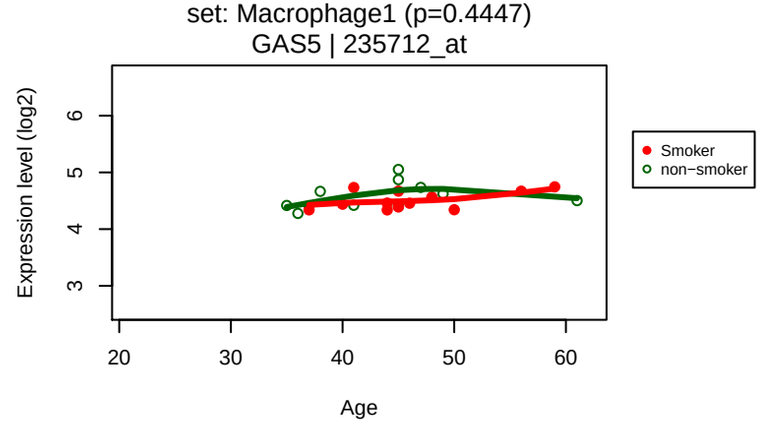

set: SmallAirway1 (p=0.1797)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker

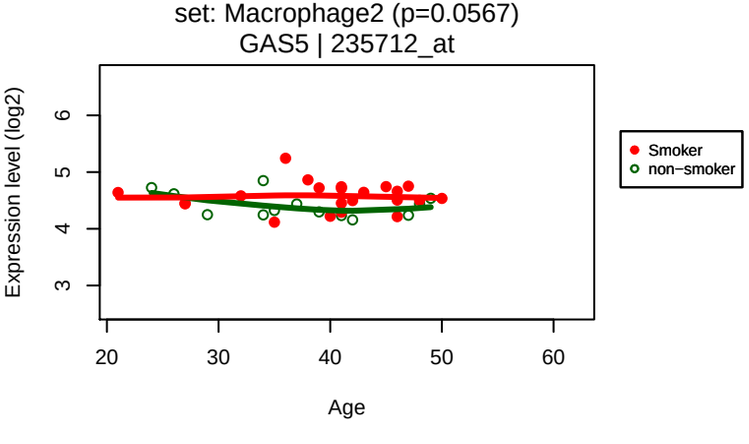

set: SmallAirway2 (p=0.5656)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker

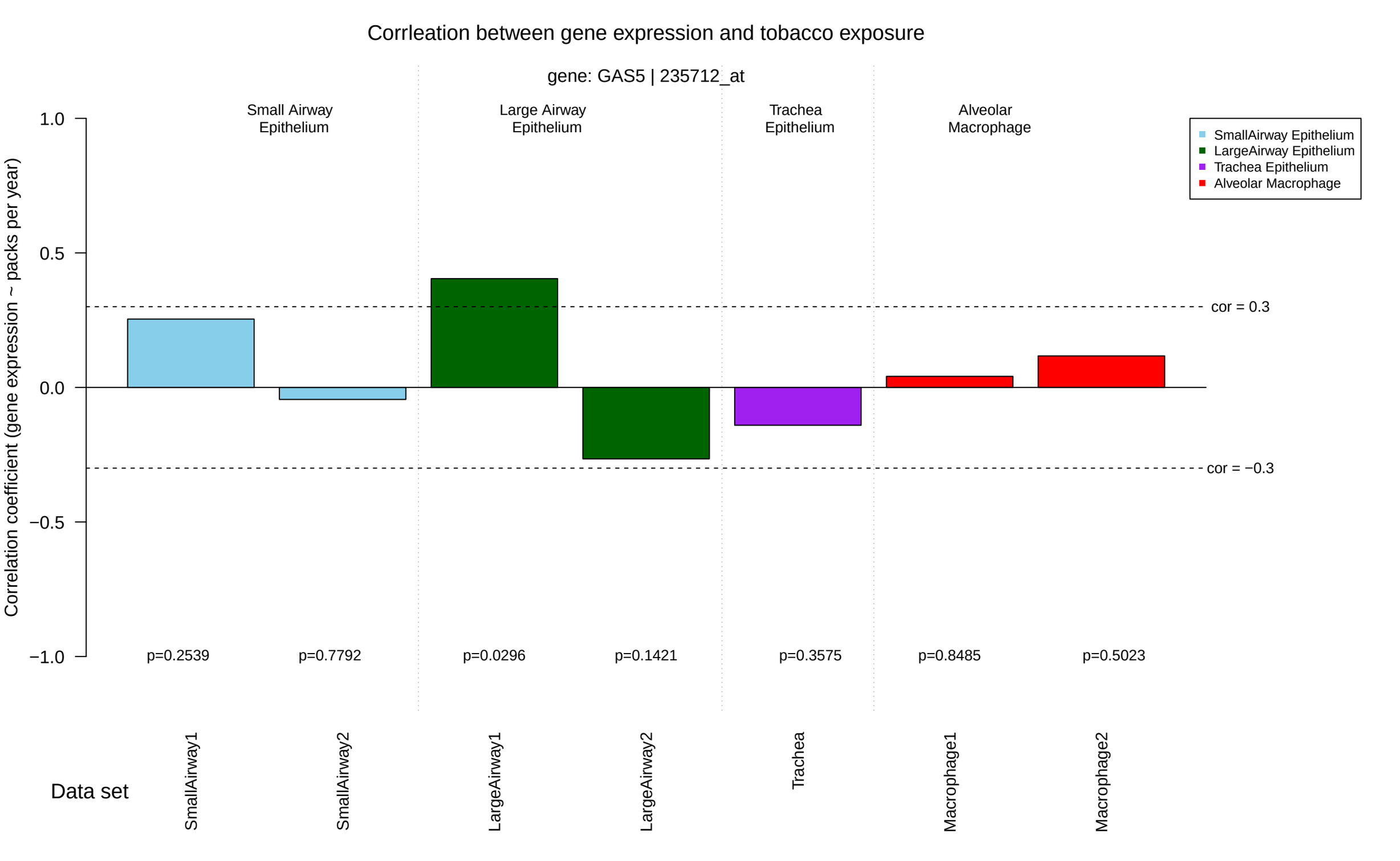

set: LargeAirway1 (p=0.012)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.5211)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0176)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.4447)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0567)
GAS5 | 235712_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: GAS5 | 235712_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2539
p=0.7792
p=0.0296
p=0.1421
p=0.3575
p=0.8485
p=0.5023
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2